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College of Engineering

CPE310: COMPUTER NETWORKS AND SECURITY

LECTURE AND LAB

LAB WORKSHEET 1

BASIC CONFIGURATION AND SET-UP OF ROUTERS AND SWITCHES

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THU | 6:00 PM – 9:00 PM

CEIT-03-602P

ENGR. DARIUS I. TRANCE

Lab Worksheet 1
Basic Configuration and Set-up of Routers and Switches

I. Objectives

To be able to familiarize the students in the use of network simulators like Cisco Packet Tracer and explore the significance of each CLI modes.
To be able to familiarize the students in basic configuration and set-up of routers and switches.

II. Discussion of the Theory

Figure 1: Sample view of Cisco Packet Tracer

Packet Tracer is a simulation tool designed by Cisco Systems. It allows users to create network topologies to simulate the configuration of routers and switches using a command line interface. Throughout the lab modules you will be configuring different commands that require changing from one mode to another.

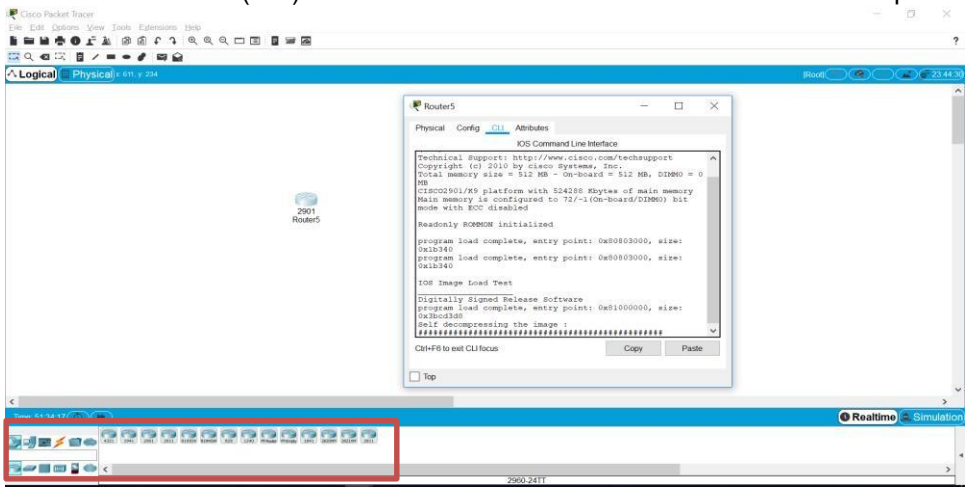
The basic CLI modes that we will be referring below are as following:

1. **Cisco>** ←User EXEC Mode
2. **Cisco#** ← Privileged EXEC mode
3. **Cisco (config)#** ←Global configuration Mode
4. **Cisco (config-if)#** ←Interface configuration Mode
5. **Cisco (config-line)#** ← Line configuration Mode

CONFIGURING YOUR ROUTER

Step 1: Getting Started

Access a router (2901 model) and place it to workspace. Double click the router and go to its command line interface (CLI). Devices are in the bottom left corner of the workspace.



Step 2: Configure an administration password (enable secret password*).*

Enter into Global Configuration mode from the Privileged EXEC mode:

Router> enable <– User EXEC Mode
Router# configure terminal <– Privileged EXEC mode
Router(config)# <– *Global Configuration Mode*

In Global Configuration Mode you configure parameters that affect the whole router device. Here we will configure the **enable secret** password that you will be using from now on to enter into Privileged EXEC Mode from User EXEC Mode.

Router(config)# enable secret *cisco123*

From now on, when you log in from user EXEC mode you will be asked for a password.

Step 3: Configure a password for telnet access and console access.

It is suggested also to configure a password for the Telnet Lines (VTY lines) which will secure your access when connecting via Telnet over the network. This goes with the console access as well.

Router(config)# line vty 0 4
Router(config-line)# password *cisco123*
Router(config-line)# login
Router(config-line)# exit
Router(config)#

Router(config)# line console 0
Router(config-line)# password *cisco123*
Router(config-line)# login
Router(config-line)# exit

Router(config)#

Step 4: Set up a Hostname to distinguish the router in the network.

To differentiate your Router from other devices in the network, you should configure a hostname to your router.

Router(config)# hostname *My-Router*
My-Router(config)#

Notice that your Router prompt changes to the new hostname that you have just set.

Step 5: **Configure IP addresses for Router Interfaces**

This is an essential step in order for your router to be able to forward packets in the network. The most basic parameter for a Router Interface is the IP address. From Global Configuration Mode you need to enter into Interface Configuration Mode:

My-Router(config)# interface gigabitethernet 0/0
My-Router(config-if)# ip address 100.100.100.1 255.255.255.252
My-Router(config-if)# no shutdown
My-Router(config-if)# exit
My-Router(config)#

My-Router(config)# interface gigabitethernet 0/1
My-Router(config-if)# ip address 192.168.10.1 255.255.255.0
My-Router(config-if)# no shutdown
My-Router(config-if)# exit
My-Router(config)#

Step 6: **Save your configuration**

Save your current running configuration into NVRAM. This will overwrite the startup configuration.

My-Router(config)# exit
My-Router# copy running-config startup-config

Step 7: Verifications

Use show commands to verify status, setting and configurations. Use the IOS help (?) You can display your current configuration to verify your settings as following: **My-Router#** show ?

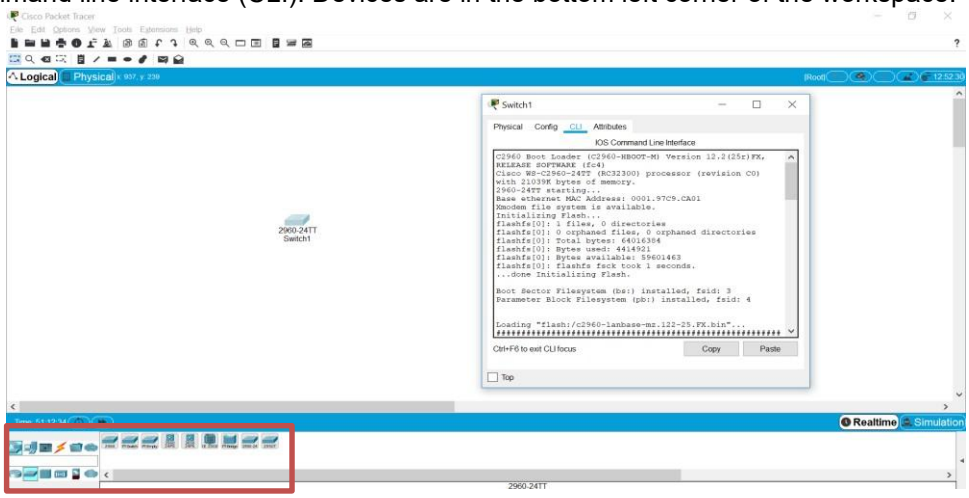
My-Router# show running-config

My-Router# show ip interface brief

CONFIGURING YOUR SWITCH

Step 1: Getting Started

Access a switch (2960 model) and place it to workspace. Double click the switch and go to its command line interface (CLI). Devices are in the bottom left corner of the workspace.



Step 2: Configure an administration password (enable secret password*).*

Switch> enable
Switch# configure terminal
Switch (config)# enable secret *cisco123*

Step 3: Configure a password for telnet access and console access.

Switch (config)# line vty 0 15
Switch (config-line)# password *cisco123*
Switch (config-line)# login
Switch (config-line)# exit
Switch (config)#

Switch (config)# line console 0
Switch (config-line)# password *cisco123*
Switch (config-line)# login
Switch (config-line)# exit
Switch (config)#

Step 4: Set up a Hostname to distinguish the switch in the network.

Switch(config)# hostname *Access-Switch1*
Access-Switch1(config)#

Step 5. Assign IP address to the Switch for Management *!Management IP is assigned to Vlan 1 by default*

Access-Switch1 (config)# interface vlan 1
Access-Switch1 (config-if)# ip address 10.1.1.200 255.255.255.0
Access-Switch1 (config-if)# exit
Access-Switch1 (config)#

Step 6. Assign default gateway to the switch

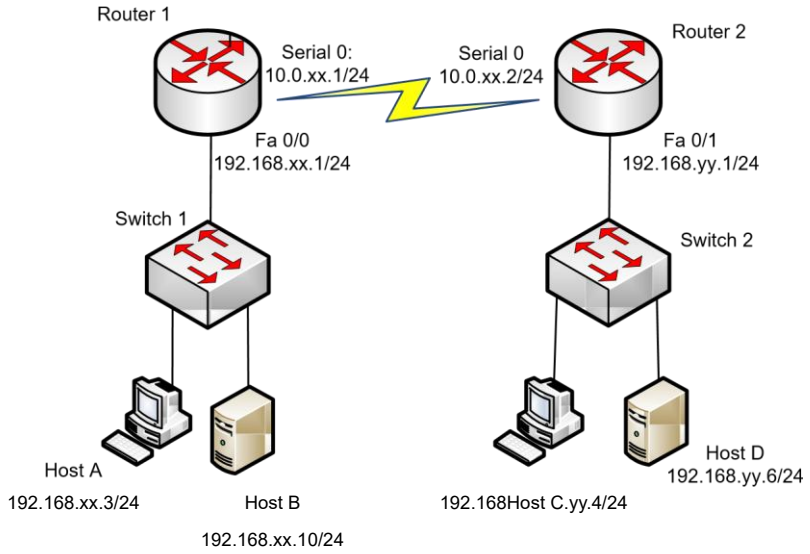
Access-Switch1 (config)# ip default-gateway 10.1.1.254

Step 7. Disable unneeded ports on the switch
! This step is optional but enhances security
! For 2960 switch, it has 24 ports. Let's try to disable the ports 13-24.

Access-Switch1 1(config)# interface range fa 0/13-24
Access-Switch1 (config-if-range)# shutdown
Access-Switch1 (config-if-range)# exit
Access-Switch1 (config)#

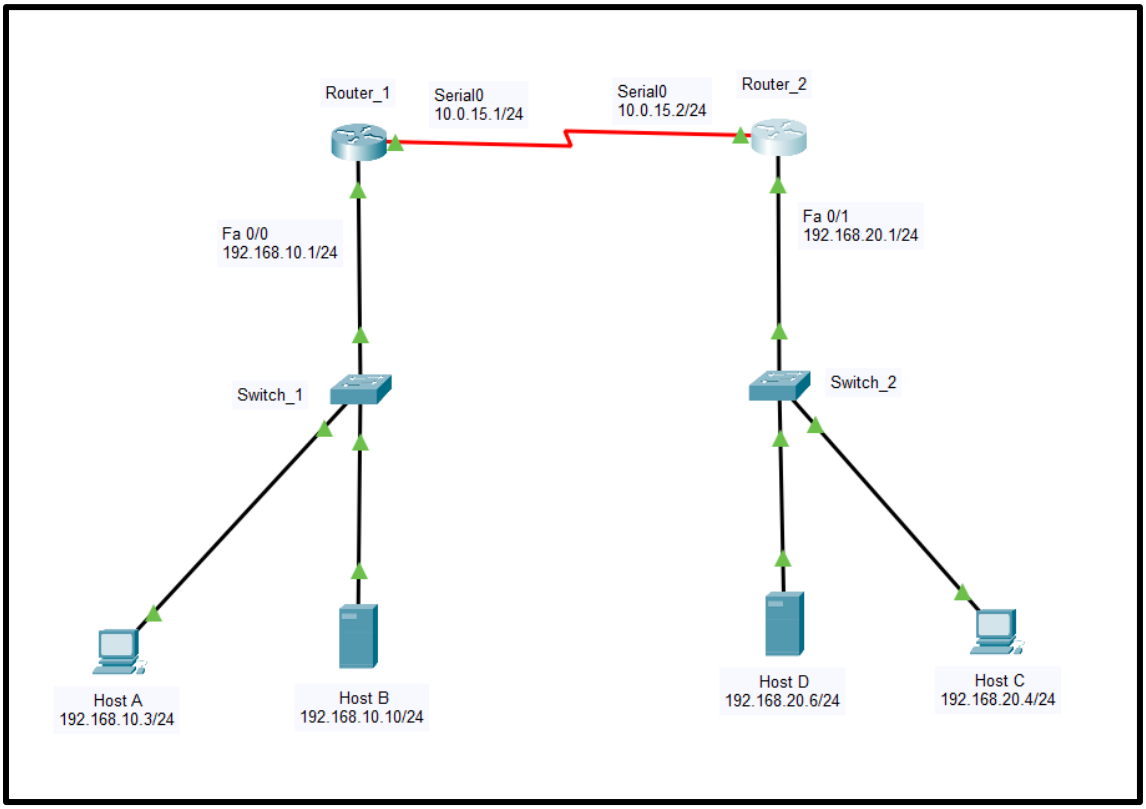
Step 8. Save the configuration
! A shortcut way of saving running configuration to start-up configuration.
Access-Switch1 (config)# wr

III. Lab Exercise



Task 1
Set-up the above network topology and save it as *Lab1 Basic Configurations Pair xx Block y*. (Example Lab 1 Basic Configuration Pair 01 Block 1)

Be mindful of the cable being used to connect the devices.



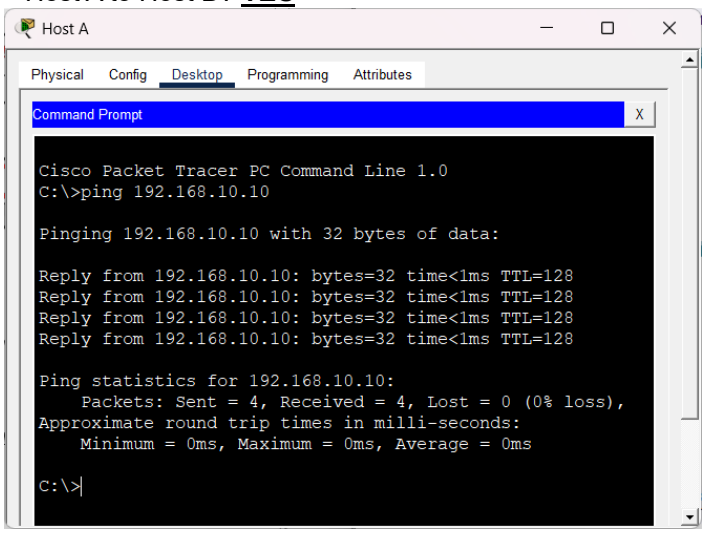
Task 2.
Configure the routers and switches with details provided on the above topology.
Named the routers as: Router 1 → Router_1
Router 2 → Router_2

Task 3.

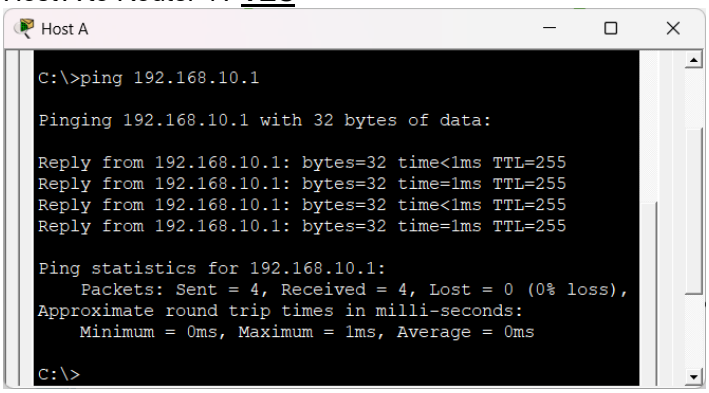
Step 1: Perform basic connectivity test from hosts to the routers.

a. Is ping successful from

Host A to Host B? ***YES***



Host A to Router 1? ***YES***



Host A to Router 2's serial 0? ***YES***

Host A

```
C:\>ping 10.0.15.2

Pinging 10.0.15.2 with 32 bytes of data:

Reply from 10.0.15.2: bytes=32 time=9ms TTL=254
Reply from 10.0.15.2: bytes=32 time=5ms TTL=254
Reply from 10.0.15.2: bytes=32 time=7ms TTL=254
Reply from 10.0.15.2: bytes=32 time=6ms TTL=254

Ping statistics for 10.0.15.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 9ms, Average = 6ms

C:\>
```

Top

Router 1 to Router 2? ***YES***

Router0

```
Router_1#ping 10.0.15.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.15.2, timeout is
2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/
max = 1/6/9 ms

Router_1#
```

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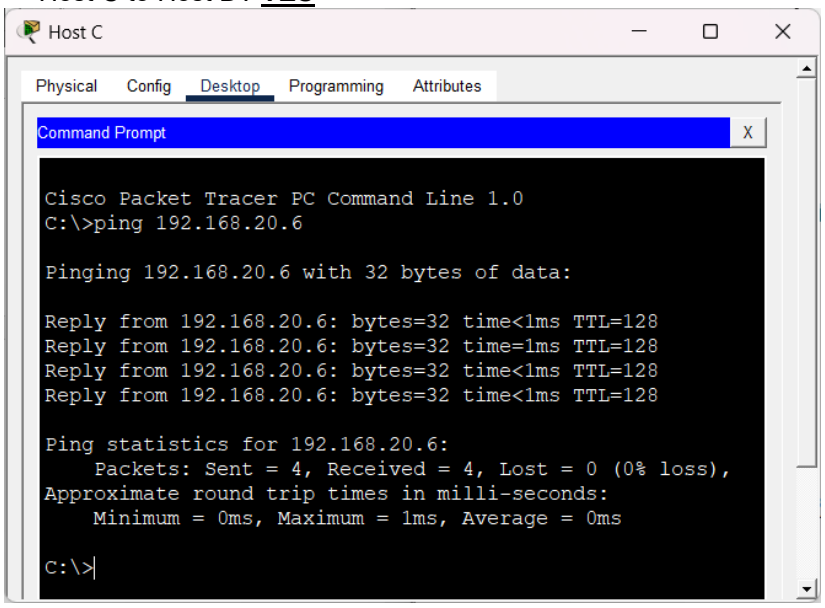
Host A to Host C? ***YES***

Host A

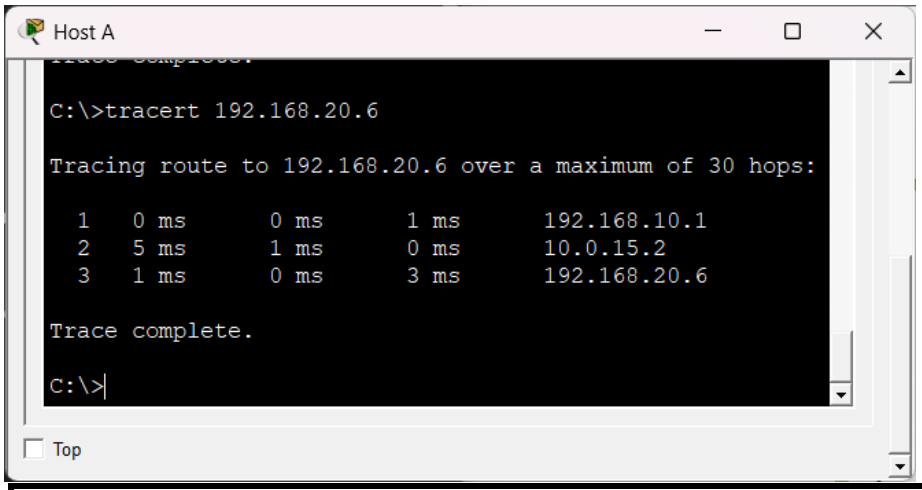
```
C:\>ping 192.168.20.4

Pinging 192.168.20.4 with 32 bytes of data:

Reply from 192.168.20.4: bytes=32 time=5ms TTL=126
Reply from 192.168.20.4: bytes=32 time=3ms TTL=126
Reply from 192.168.20.4: bytes=32 time=3ms TTL=126
Reply from 192.168.20.4: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\>
```

Top

Host C to Host D? ***YES***

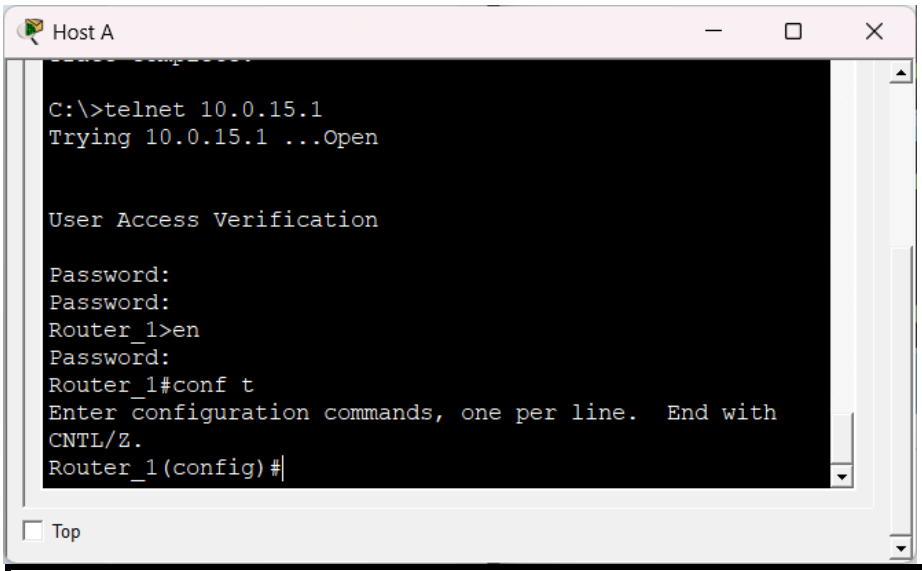


b. Perform a trace from Host A to Host D, what is your observation on the result? Use the **tracert** command.



The trace from **Host A (192.168.10.3)** to **Host D (192.168.20.6)** was successful. The packet passed through the default gateway and the inter-router link, then reached Host D, which shows routing between the networks is working correctly.

Step 2: Telnet either from Host A to Router 1 or Host C to Router 2. What is your observation on the result? Use the **telnet** command.



The Telnet session from **Host A** to **Router_1** worked as expected. I connected to Router_1 using its IP address, was prompted for the VTY password to gain user access, then entered privileged mode using the enable password and reached global configuration mode. The VTY password exists to protect remote Telnet access, and without configuring the VTY lines with a password and login, the router would not allow any Telnet connection at all.

Task 4.
Using **show** commands, verify the following;

1. **show version**

IOS version of Router 1:
- **Version 15.1 (4) m4**

IOS image of Router 1:
- **System image file is "flash0:c2800nm-advipservicesk9-mz.151-4.M4.bin"**

uptime of Switch 2:
- **Switch uptime is 39 minutes**

RAM of Router 2:
- **cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory**

Configuration Register of Router 2:
- **Configuration register is 0x2102**

2. **show interface Serial 0**
Interface status in Router 1:

- **Serial0/0/0 is up, line protocol is up (connected)**

Bandwidth:

- **BW 1544 Kbit**

Reliability :

- **reliability 255/255**

MTU:

- **MTU 1500 bytes**

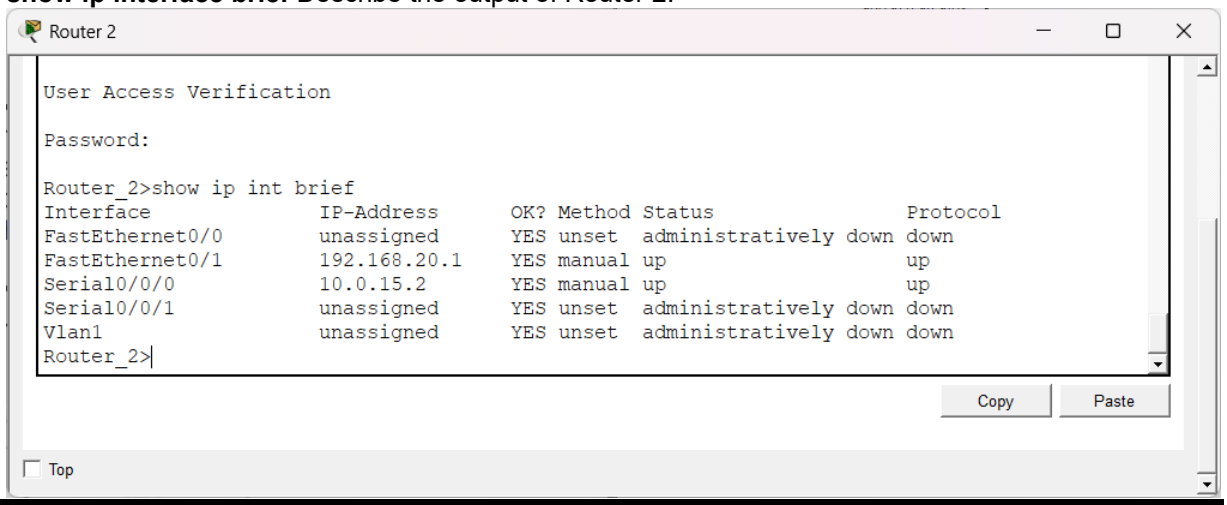
Output drops:

- **Total output drops: 0**

Duplex:

- **Serial interfaces do not use duplex settings.**

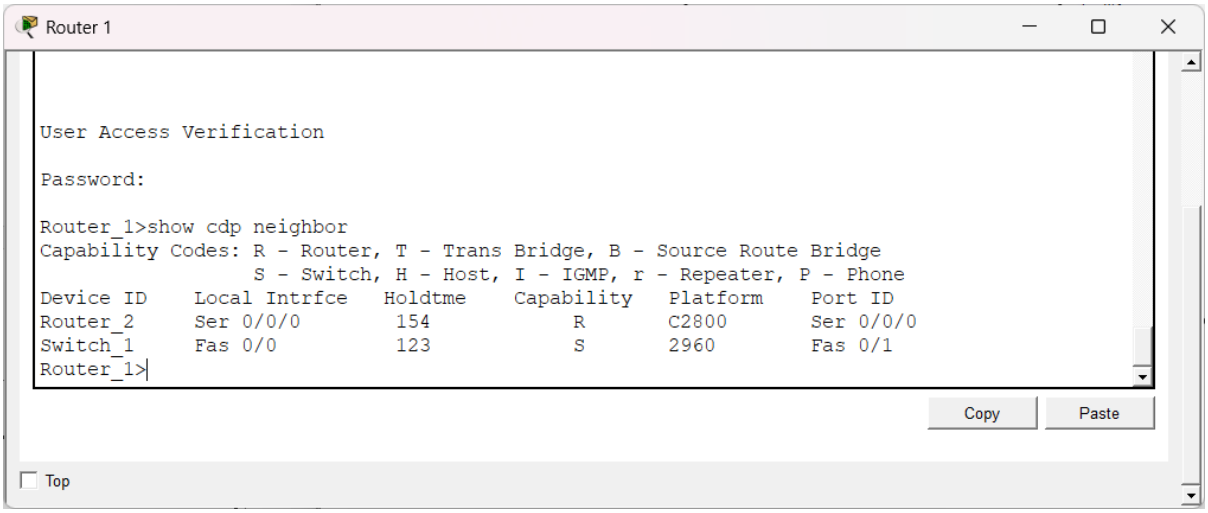
3. **show ip interface brief** Describe the output of Router 2:



The output shows that Router_2 has two active interfaces and several inactive ones. **Fa0/1 (192.168.20.1)** is up and running, which indicates it is connected to the LAN and operational. **Serial0/0/0 (10.0.15.2)** is also up, confirming a working serial link to another router. **Fa0/0**, **Serial0/0/1**, and **Vlan1** are administratively down, meaning they are disabled and not participating in network communication.

4. **show cdp neighbor**

Describe the output in Router 1



The output shows that Router_1 has two directly connected neighbors discovered through CDP. **Router_2** is connected to **Router_1** via the **Serial0/0/0** interface on both devices, confirming a router to router serial link. **Switch_1** is connected to **Router_1** through **FastEthernet0/0** on the router and **FastEthernet0/1** on the switch, confirming the LAN connection between the router and the switch.

5. **show run**

Describe the output in Router 2

The running configuration of Router_2 indicates that two interfaces are enabled and in use. **FastEthernet0/1** is assigned an IP address **192.168.20.1/24** and handles communication for the local LAN, while **Serial0/0/0** is configured with the IP address **10.0.15.2/30** and connects **Router_2** to **Router_1** over the WAN link. Both interfaces are active and operating normally. The other interfaces, including **FastEthernet0/0**, **Serial0/0/1**, and **Vlan1**, are disabled and not involved in network operations.

The configuration also shows that the console and VTY lines are secured with passwords and login is enabled, allowing local and remote access to the router through Telnet.

Router 2

Physical Config CLI Attributes

IOS Command Line Interface

```
interface FastEthernet0/0
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface FastEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 10.0.15.2 255.255.255.252
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.10.0 255.255.255.0 10.0.15.1
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 password romar123
 login
!
line aux 0
!
line vty 0 4
 password romar123
 login
!
!
!
```

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IV. Conclusion

In this activity, I was able to build and configure a simple network made up of two routers, two switches, and four host devices. I learned how correct IP addressing allows devices on different networks to communicate properly, and how serial connections between routers need the correct hardware module and configuration to function. I also learned how CDP helps identify directly connected devices and confirm physical connections within the topology. Another key takeaway was the need to configure administrative access, such as enable secrets and VTY passwords, so remote management through Telnet works. Without these settings, remote access to the router fails. By using commands like ping, tracert, telnet, show ip interface brief, show version, I was able to test connectivity, check interface status, and verify overall router and switch configuration. This activity showed how both correct physical setup and proper configuration are required to create a working and manageable network.

End of Lab Worksheet 1

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CPE310: COMPUTER NETWORKS AND SECURITY

LECTURE AND LAB

LAB WORKSHEET 2

BASIC SWITCH CONFIGURATION

HINAMPAS, ROMAR

THU | 6:00 PM – 9:00 PM

CEIT-03-602P

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Lab Worksheet 2
Basic Switch Configuration

I. Objective

In this lab, you will examine and configure a standalone LAN switch. Although a switch performs basic functions in its default out-of-the-box condition, there are a number of parameters that a network administrator should modify to ensure a secure and optimized LAN. This lab introduces you to the basics of switch configuration.

II. Discussion of the Theory

Recall that with shared Ethernet networks using hubs, many hosts are connected to a single broadcast and collision domain. In other words, shared Ethernet media operates at OSI Layer 1.

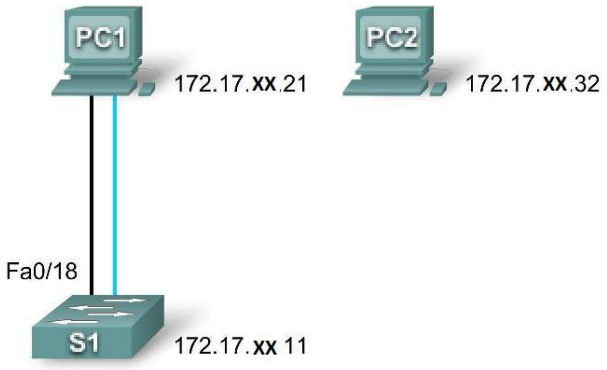
Each host must share the available bandwidth with every other connected host. When more than one host tries to talk at one time, a collision occurs, and everyone must back off and wait to talk again. This forces every host to operate in half-duplex mode, by either talking *or* listening at any given time. In addition, when one host sends a frame, all connected hosts hear it. When one host generates a frame with errors, everyone hears that, too.

At its most basic level, an Ethernet switch provides isolation from other connected hosts in several ways:

- The collision domain's scope is severely limited. On each switch port, the collision domain consists of the switch port itself and the devices directly connected to that port—either a single host or if a shared-media hub is connected, the set of hosts connected to the hub.
- Host connections can operate in full-duplex mode because there is no contention on the media. Hosts can talk *and* listen at the same time.
- Bandwidth is no longer shared. Instead, each switch port offers dedicated bandwidth across a switching fabric to another switch port. (These connections change dynamically.)
- Errors in frames are not propagated. Each frame received on a switch port is checked for errors. Good frames are regenerated when they are forwarded or transmitted. This is known as *storeand-forward* switching technology, where packets are received, stored for inspection, and then forwarded.
- You can limit broadcast traffic to a volume threshold.
- Other types of intelligent filtering or forwarding become possible.

The switch builds and maintains a table, which is called a MAC table. This table matches a destination MAC address with the port that is used to connect to a node. For each incoming frame, the destination MAC address in the frame header is compared to the list of addresses in the MAC table. Switches then use MAC addresses as they decide whether to filter, forward, or flood frames.

III. Lab Exercise



where: xx – pair number

TASK 1: VERIFY THE DEFAULT SWITCH CONFIGURATION

Switch>enable
Switch#

Notice that the prompt changed in the configuration to reflect privileged EXEC mode.

Step 1: Examine the current switch configuration.

Examine the current running configuration file.

Switch# show running-config

How many Fast Ethernet interfaces does the switch have? **24 FastEthernet Interfaces**
How many Gigabit Ethernet interfaces does the switch have? **2 GigabitEthernet Interfaces**
What is the range of values shown for the vty lines? **line vty 0 4 and line vty 5 15, therefore, 16 total vty lines**

Step 2: Examine the current contents of NVRAM:

Switch# show startup-config

startup-config is not present
Why does the switch give this response?
The switch says "startup-config is not present" because the switch configuration has not been saved to NVRAM yet.

Step 3: Examine the characteristics of the virtual interface VLAN1:

Switch# show interface vlan1

Is there an IP address set on the switch? **No**
What is the MAC address of this virtual switch interface? **00d0.d3d2.6407**
Is this interface up? **No, it shows "Vlan1 is administratively down, line protocol is down"**
Now view the IP properties of the interface:

Switch#show ip interface vlan1

What output do you see?
Vlan1 is administratively down, line protocol is down
Internet protocol processing disabled

Step 4: Display Cisco IOS information.
Examine the following version information that the switch reports.

Switch#show version

What is the Cisco IOS version that the switch is running? **Version 15.0(2)SE4**
What is the system image filename? **flash:c2960-lanbasek9-mz.150-2.SE4.bin**
What is the base MAC address of this switch? **00:D0:D3:D2:64:07**

Step 5: Examine the Fast Ethernet interfaces.
Examine the default properties of the Fast Ethernet interface used by PC1.

Switch#show interface fastethernet 0/18

Is the interface up or down? **The interface is up. It shows "FastEthernet0/18 is up, line protocol is up (connected)"**
What event would make an interface go up? **An interface would go up when a device such as PC is connected with a working cable and the port is enabled (not shutdown).**
What is the MAC address of the interface? **0001.64a1.6612**
What is the speed and duplex setting of the interface? **Full-duplex, 100Mb/s**

Step 6: Examine VLAN information.
Examine the default VLAN settings of the switch.

Switch#show vlan

What is the name of VLAN 1? **default**
Which ports are in this VLAN? **All ports are in VLAN 1 (Fa0/1 to Fa0/24 and Gig0/1 and Gig0/2**
Is VLAN 1 active? **Yes, it shows status: active**
What type of VLAN is the default VLAN? **it says enet, which means Ethernet VLAN**

TASK 2: CREATE A BASIC SWITCH CONFIGURATION

Step 1: Assign a name to the switch.
In the last step of the previous task, you configured the hostname.
Here's a review of the commands used.
Switch#config t *!shortcut for configure terminal*
Swicth (config)#hostname *S1*
S1(config)#exit
S1#

Step 2: Set the access passwords.
Enter config-line mode for the console. Set the login password to *cisco*.
Also configure the vty lines 0 to 15 with the password *cisco*.

S1#configure terminal

Enter the configuration commands, one for each line. When you are finished, return to global configuration mode by entering the exit command or pressing Ctrl-Z.

S1(config)#line console 0
S1(config-line)#password *cisco*
S1(config-line)#login
S1(config-line)#line vty 0 15
S1(config-line)#password *cisco*
S1(config-line)#login
S1(config-line)#exit

Why is the login command required?

The login command makes the switch require a password when someone connects to the console or VTY line. Without it, the password I set will not be enforced.

Step 3: Set the command mode passwords.

Set the enable secret password to *class*. This password protects access to privileged EXEC mode.

S1(config)#enable secret *class*

Step 4: Configure the Management IP address of the switch.

Before you can manage S1 remotely from PC1, you need to assign the switch an IP address. The default configuration on the switch is to have the management of the switch controlled through VLAN 1.

However, a best practice for basic switch configuration is to change the management VLAN to a VLAN other than VLAN 1. The selection of VLAN xx is arbitrary and in no way implies you should always use VLAN xx.

First, you will create the new VLAN xx on the switch. Then you will set the IP address of the switch to 172.17.xx.11 with a subnet mask of 255.255.255.0 on the internal virtual interface VLAN xx.

S1(config)#vlan *xx*
S1(config-vlan)#exit
S1(config)#interface vlan*xx*
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlanxx, changed state to down
S1(config-if)#ip address 172.17. *xx*.11 255.255.255.0
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#

Notice that the VLAN xx interface is in the down state even though you entered the command no shutdown. The interface is currently down because no switch ports are assigned to VLAN xx. Assign all user ports to VLAN xx.

S1(config)#interface range fa0/1 – 24
S1(config-if-range)#switchport access vlan *xx*
S1(config-if-range)#exit
S1(config-if-range)#

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlanxx, changed state to up

Notice in the above output that VLAN 1 interface goes down because none of the ports are assigned to VLAN 1. After a few seconds, VLAN xx will come up because at least one port is now assigned to VLAN xx.

Step 5: Set the switch default gateway.

S1 is a layer 2 switch, so it makes forwarding decisions based on the Layer 2 header. If multiple networks are connected to a switch, you need to specify how the switch forwards the internetwork frames, because the path must be determined at Layer 3. This is done by specifying a default gateway address that points to a router or Layer 3 switch. Although this activity does not include an external IP gateway, assume that you will eventually connect the LAN to a router for external access. Assuming that the LAN interface on the router is 172.17.xx.1, set the default gateway for the switch.

S1(config)#ip default-gateway 172.17. *xx*.1 **S1(config)#**exit

Step 6: Verify the management LANs settings.

Verify the interface settings on VLAN xx.

S1#show interface vlan *xx*

What is the bandwidth on this interface? **BW 100000 Kbit**
What are the VLAN states? VLAN1 is **up** Line protocol is **up**
What is the queuing strategy? **FIFO (First In First Out)**

Step 7: Configure the IP address and default gateway for PC1.

Set the IP address of PC1 to 172.17.xx.21, with a subnet mask of 255.255.255.0. Configure a default gateway of 172.17.xx.11

Step 8: Verify connectivity.

To verify the host and switch are correctly configured, ping the IP address of the switch (172.17.xx.11) from PC1.
Was the ping successful? **Yes**
If not, troubleshoot the switch and host configuration. Note that this may take a couple of tries for the pings to succeed.

Step 9: Configure the port speed and duplex settings for a Fast Ethernet interface.

Configure the duplex and speed settings on Fast Ethernet 0/18. Use the end command to return to privileged EXEC mode when finished.

S1#configure terminal
S1(config)#interface fastethernet 0/18
S1(config-if)#speed 100
S1(config-if)#duplex full
S1(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/18, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlanxx, changed state to down
%LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to down
%LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/18, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlanxx, changed state to up

The line protocol for both interface FastEthernet 0/18 and interface VLAN xx will temporarily go down. The default on the Ethernet interface of the switch is auto-sensing, so it automatically negotiates optimal settings. You should set duplex and speed manually only if a port must

operate at a certain speed and duplex mode. Manually configuring ports can lead to duplex mismatches, which can significantly degrade performance.
Verify the new duplex and speed settings on the Fast Ethernet interface.

S1#show interface fastethernet 0/18

Step 10: Save the configuration.

You have completed the basic configuration of the switch. Now back up the running configuration file to NVRAM to ensure that the changes made will not be lost if the system is rebooted or loses power.

S1#copy running-config startup-config
Destination filename [startup-config]?[Enter] Building configuration... [OK]
S1#

TASK 3: MANAGING THE MAC ADDRESS TABLE

Step 1: Record the MAC addresses of the hosts.

Determine and record the Layer 2 (physical) addresses of the PC network interface cards using the following commands:
Click the PC > Device Tab > Command prompt > type **ipconfig /all**
PC1: **0040.0B8D.B619**
PC2: **0001.9691.E938**

Step 2: Determine the MAC addresses that the switch has learned.

Display the MAC addresses using the show mac-address-table command in privileged EXEC mode.

S1#show mac-address-table

How many dynamic addresses are there? **There is only one dynamic address (VLAN 10 - 0040.0b8d.b619)**
How many MAC addresses are there in total? **Only one - 0040.0b8d.b619 (PC1)**
Do the dynamic MAC addresses match the host MAC addresses? **Yes, it matched PC1**

Step 3: List the show mac-address-table options.

S1#show mac-address-table ?

How many options are available for the show mac-address-table command? **There are 3 options: dynamic dynamic entry type**
interfaces interface entry type
static static entry type
Show only the MAC addresses from the table that were learned dynamically.

S1#show mac-address-table address <PC1 MAC here>

How many dynamic addresses are there? **Only one dynamic address**

Step 4: Clear the MAC address table.

To remove the existing MAC addresses, use the clear mac-address-table command from privileged EXEC mode.

S1#clear mac-address-table dynamic

Step 5: Verify the results.

Verify that the MAC address table was cleared.

S1#show mac-address-table

How many static MAC addresses are there? **None**
How many dynamic addresses are there? **None**

Step 6: Examine the MAC table again.

More than likely, an application running on your PC1 has already sent a frame out the NIC to S1. Look at the MAC address table again in privileged EXEC mode to see if S1 has relearned the MAC address for PC1

S1#show mac-address-table

How many dynamic addresses are there? **There is only one.**
Why did this change from the last display? **At first, there's still none shown but after I ping PC1 from S1, it shows one dynamic address. This happened because by pinging either way, the switch dynamically learned MAC addresses.**

If S1 has not yet relearned the MAC address for PC1, ping the VLAN xx IP address of the switch from PC1

Step 7: Set up a static MAC address.

To specify which ports a host can connect to, one option is to create a static mapping of the host MAC address to a port. Set up a static MAC address on Fast Ethernet interface 0/18 using the address that was recorded for PC1 in Step 1 of this task.

S1(config)#mac-address-table static <PC1 MAC> vlan xx interface fastethernet0/18

Step 8: Verify the results.

Verify the MAC address table entries.
S1#show mac-address-table
How many total MAC addresses are there? **There is only one MAC address.**
How many static addresses are there? **Only one static MAC address**

TASK 4: CONFIGURING PORT SECURITY

Step 1: Configure a second host.

A second host is needed for this task. Set the IP address of PC2 to 172.17.xx.32, with a subnet mask of 255.255.255.0 and a default gateway of 172.17.xx.11. Do not connect this PC to the switch yet.

Step 2: Verify connectivity.

Verify that PC1 and the switch are still correctly configured by pinging the VLANxx IP address of the switch from the host.
Were the pings successful? **Yes**
If the answer is no, troubleshoot the host and switch configurations.

Step 3: Copy the host MAC addresses.
Write down the MAC addresses from Task 3
PC1: **0040.0B8D.B619**
PC2: **0001.9691.E938**

Step 4: Determine which MAC addresses that the switch has learned.
Display the learned MAC addresses using the show mac-address-table command in privileged EXEC mode.

S1#show mac-address-table

How many dynamic addresses are there? **There is only one dynamic address**
Do the MAC addresses match the host MAC addresses? **Yes**

Step 5: List the port security options.
Explore the options for setting port security on interface Fast Ethernet 0/18.

S1# configure terminal
S1(config)#interface fastethernet 0/18

S1(config-if)#switchport port-security **?**

aging	Port-security aging commands
mac-address	Secure mac address
maximum	Max secure addresses
violation	Security violation mode
<cr>	

S1(config-if)# switchport port-security

Step 6: Configure port security on an access port.
Configure switch port Fast Ethernet 0/18 to accept only two devices, to learn the MAC addresses of those devices dynamically, and to block traffic from invalid hosts if a violation occurs.

S1(config-if)#switchport mode access
S1(config-if)#switchport port-security
S1(config-if)#switchport port-security maximum 2
S1(config-if)#switchport port-security mac-address sticky
S1(config-if)#switchport port-security violation protect
S1(config-if)#exit

Step 7: Verify the results.
Show the port security settings.

S1#show port-security

How many secure addresses are allowed on Fast Ethernet 0/18? **There are 2 maximum secure address on Fa0/18.**
What is the security action for this port? **The security action for this port is "protect".**

Step 8: Modify the port security settings on a port.
On interface Fast Ethernet 0/18, change the port security maximum MAC address count to 1 and to shutdown if a violation occurs.

S1(config-if)#switchport port-security maximum 1
S1(config-if)#switchport port-security violation shutdown

Step 9: Verify the results.
Show the port security settings.

S1#show port-security

Have the port security settings changed to reflect the modifications in Step 6? **Yes, the maximum secure address is now set to '1' and the security action is now set to 'Shutdown'.**
Ping the VLAN xx address of the switch from PC1 to verify connectivity and to refresh the MAC address table. You should now see the MAC address for PC1 "stuck" to the running configuration.

Step 10: Introduce a rogue host.
Disconnect PC1 and connect PC2 to port Fast Ethernet 0/18. Ping the VLAN xx address 172.17.xx.11 from the new host.
Record any observations:
After disconnecting PC1 and connecting PC2 to port Fa0/18. Then, pinging the VLAN 10 address 172.17.10.11 from PC2, the result shows "Request time out", that means the ping request was unsuccessful. This happened because the port is configured to allow only 1 secure MAC address, and a sticky MAC address of PC1 is already stuck, the switch detected a port security violation. Since the violation mode is set to 'shutdown', the switch automatically places Fa0/18 into an 'err-disable state'. The port becomes down, and communication through port Fa0/18 stops completely.

S1 CLI:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to down
%PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/18, putting Fa0/18 in err-disable state.
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address 0001.9691.E938 on port
FastEthernet0/18.
```

Step 11: Show port configuration information.
To see the configuration information for just Fast Ethernet port 0/18, issue the following command in privileged EXEC mode:

S1#show interface fastethernet 0/18

What is the state of this interface?
Fast Ethernet0/18 is **down** Line protocol is **down** **(err-disabled)**

Step 12: Reactivate the port.

If a security violation occurs and the port is shut down, you can use the no shutdown command to reactivate it. However, as long as the rogue host is attached to Fast Ethernet 0/18, any traffic from the host disables the port. Reconnect PC1 to Fast Ethernet 0/18, and enter the following commands on the switch:

S1# configure terminal
S1(config)#interface fastethernet 0/18
S1(config-if)# no shutdown
S1(config-if)#exit

*Note: Some IOS version may require a manual **shutdown** command before entering the **no shutdown** command*

IV. Conclusion

In this activity, I learned how to check the default configuration of a switch and understand its basic settings like interfaces, VLAN, IOS version, and MAC address. I also learned how to create a basic switch configuration by setting hostname, passwords, management IP address, default gateway, and configuring port speed and duplex.

After that, I managed the MAC address table by viewing, clearing, and setting a static MAC address. I saw how the switch dynamically learns MAC addresses when devices send traffic.

In the last part, I configured port security to limit the number of devices connected to a port. When I connected a different PC, the port was automatically disabled because of a security violation. This showed me how port security helps protect the network from unauthorized devices.

Overall, this activity helped me understand how a switch works, how it learns MAC addresses, and how to secure a port properly.

End of Lab Worksheet 2

CPE310: COMPUTER NETWORKS AND SECURITY

LECTURE AND LAB

PACKET TRACER WARM-UP LABS

LABORATORY 1: BASIC NETWORK CONNECTIVITY AND CONFIGURATION

HINAMPAS, ROMAR

THU | 6:00 PM – 9:00 PM

CEIT-03-602P

ENGR. DARIUS I. TRANCE

Packet Tracer Warm-Up Labs

Laboratory 1: Basic Network Connectivity and Configuration

Overview

This laboratory exercise is the first hands-on networking activity of the course.
It introduces students to basic network connectivity, starting from direct PC-to-PC connections, progressing to switch-based networks, and finally router-based inter-networking.

The activities are designed to help students:

- Understand how devices communicate
- Learn basic IP addressing
- Use the ping command for testing connectivity
- Apply basic troubleshooting skills

All activities are performed using **Cisco Packet Tracer**, a safe simulation tool widely used in networking education.

Activity 1: PC-to-PC Direct Connection

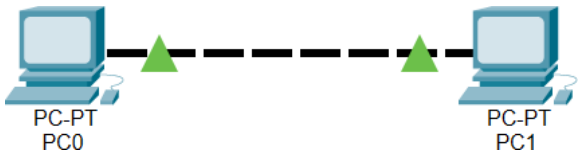
Goal: Allow two PCs to communicate and successfully ping each other via a direct connection.

Devices:

- 2 PCs
- Copper Cross-Over Cable

Steps:

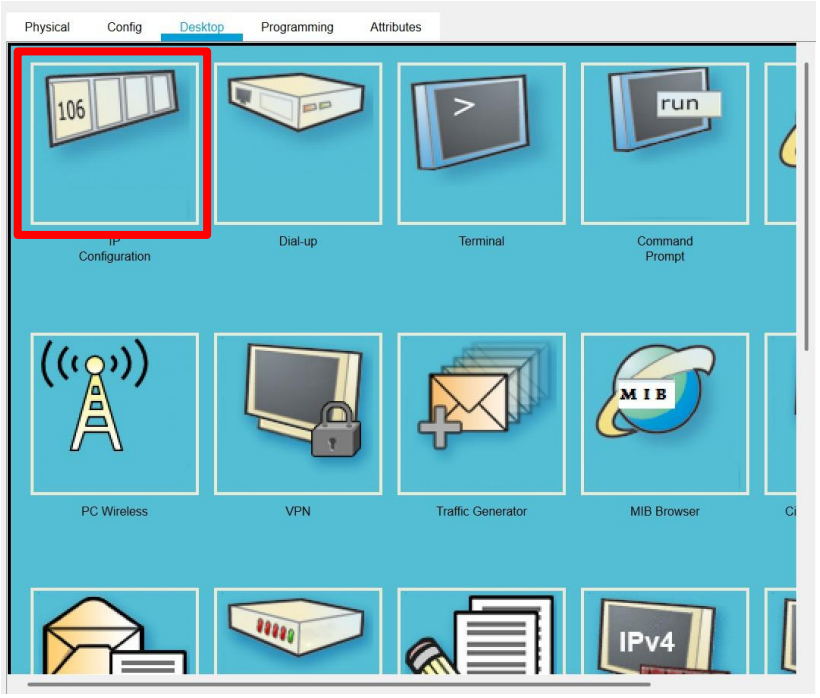
1. Open Cisco Packet Tracer.
2. From End Devices, drag 2 PCs into the workspace.
3. Click Connections → select Copper Cross-Over Cable.
4. Connect: PC0 FastEthernet0 → PC1 FastEthernet0

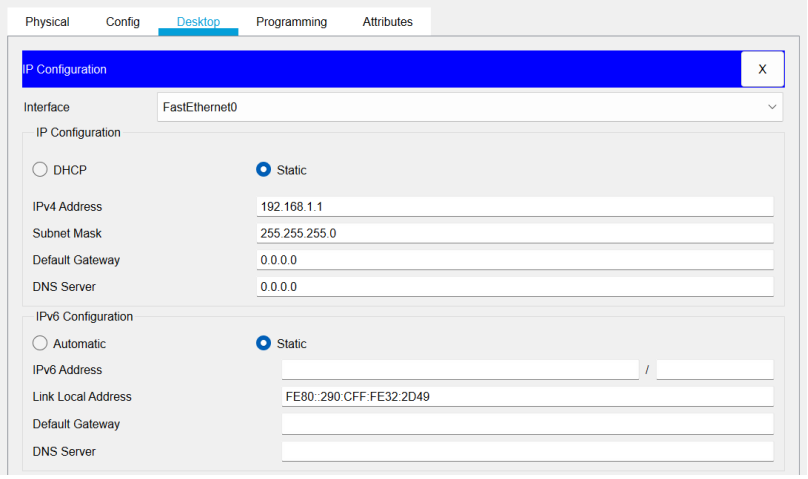


5. Configure IP addresses:
6. PC0 → Desktop → IP Configuration

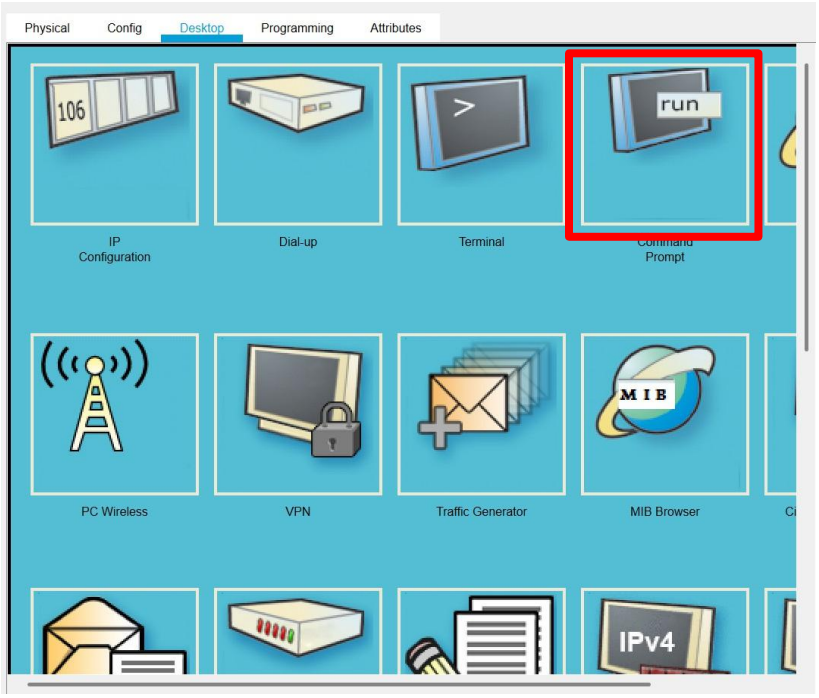
IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0

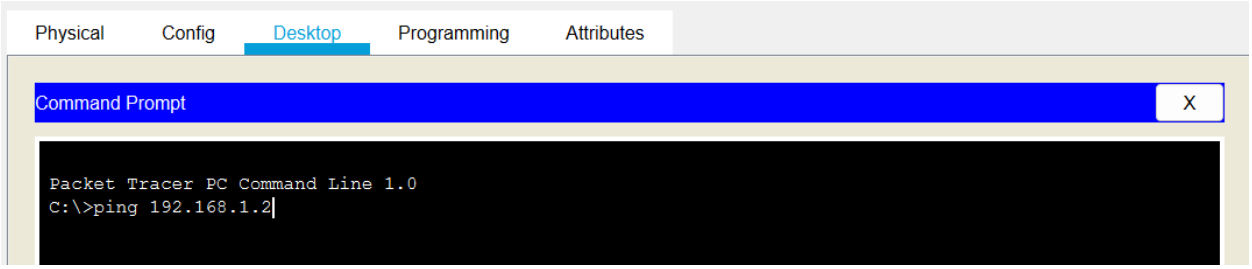
PC1 → Desktop → IP Configuration
IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0





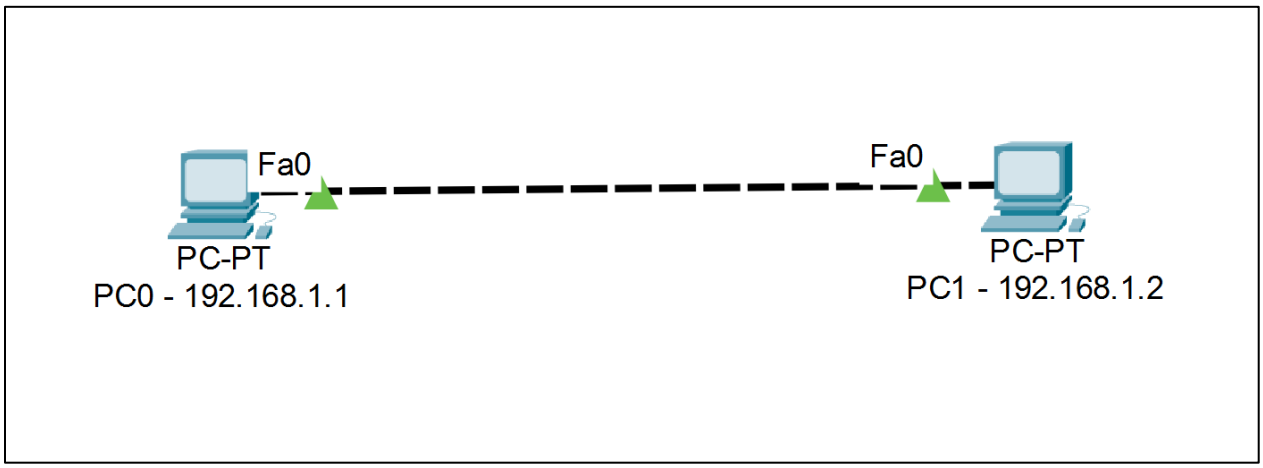
7. Test connectivity:
   PC0 → Desktop → Command Prompt
   Type:
   ping 192.168.1.2



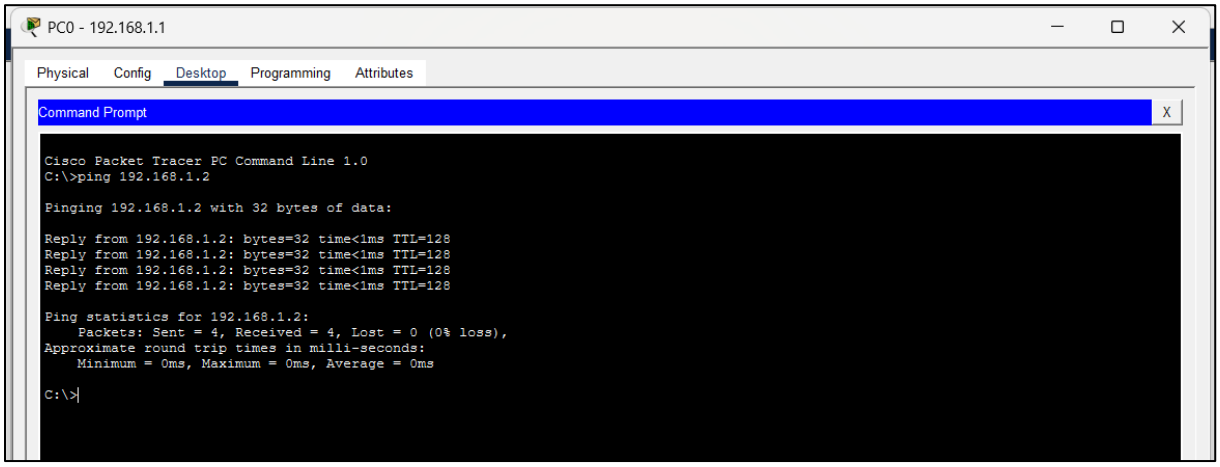


Troubleshooting:
Check if both PCs are powered ON
Verify correct cable type (Cross-Over)
Ensure both PCs are in the same subnet

Documentation:
Screenshot of network topology



Screenshot of successful ping



GUIDE QUESTIONS

1. Why is a cross-over cable used for PC-to-PC connection?

   Because it allows two similar devices (PC to PC) to directly communicate by crossing the transmit and receive signals without the need for a switch or hub.

2. What happens if the PCs have different subnet masks?

   If PCs have different subnet masks, they may not recognize or communicate with each other because they are on different networks.

3. What is the purpose of the ping command?

   The ping command is used to test network connectivity by sending packets to another device and checking if a response is received, confirming whether the connection is working properly.

Activity 2: PC-to-PC via Switch

Goal: Use a switch to allow two PCs to communicate with each other.

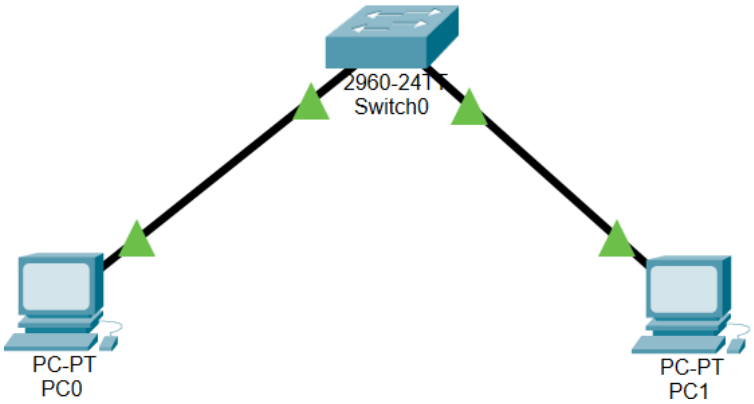
Devices:

- 2 PCs
- 1 Switch
- 2 Copper Straight-Through Cables

Steps:

1. Drag **2 PCs** and **1 switch** into the workspace.
2. Connect:

    - PC0 → Switch **Fa0/1**
    - PC1 → Switch **Fa0/2**



3. Configure IP addresses:

    - PC0: 192.168.1.1 / 255.255.255.0
    - PC1: 192.168.1.2 / 255.255.255.0

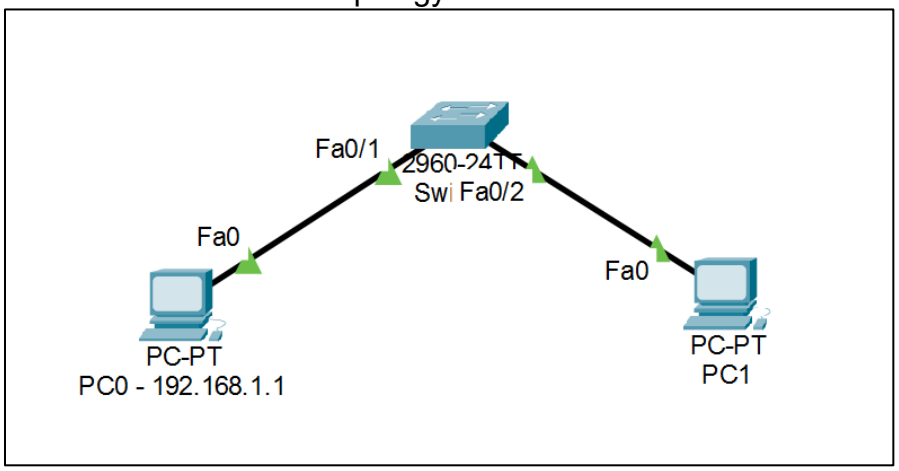
4. Perform a ping test from PC0 to PC1.

Troubleshooting:

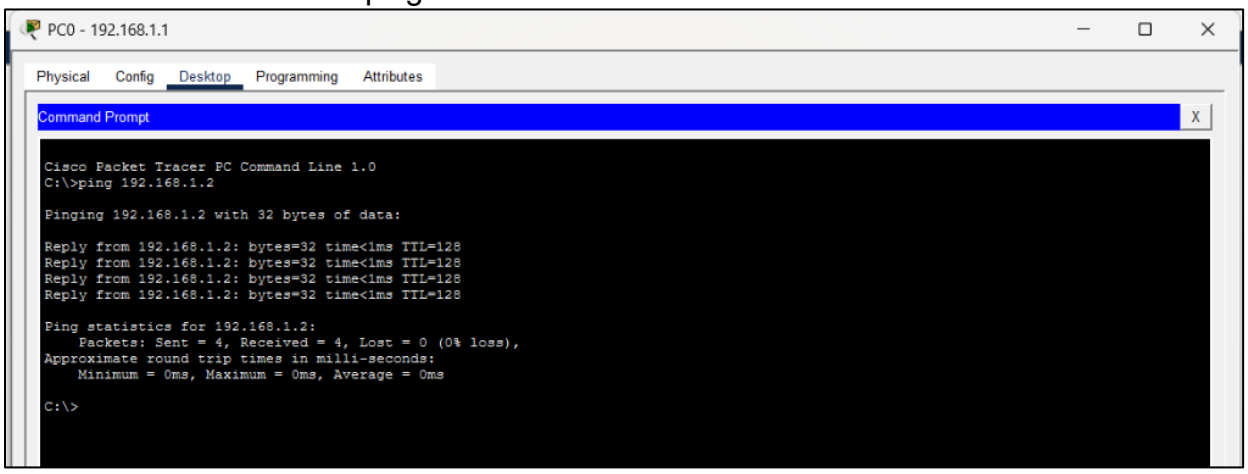
Check if switch is powered ON
Ensure both PCs belong to VLAN 1 (default)
Verify that switch ports are not disabled

Documentation:

Screenshot of network topology



Screenshot of successful ping



GUIDE QUESTIONS

1. What is the main function of a switch?
   The main function of a switch is to connect multiple devices in a network and forward data only to the correct destination device.

2. Why do switches use MAC addresses?
   Switches use MAC addresses to identify devices and determine exactly where to send data frames within a local network.

3. What is the default VLAN of a switch?
   The default VLAN of a switch is VLAN 1 because it is preconfigured by the manufacturer to allow all switch ports to communicate immediately without any additional setup.

Activity 3: Inter-networking using Router

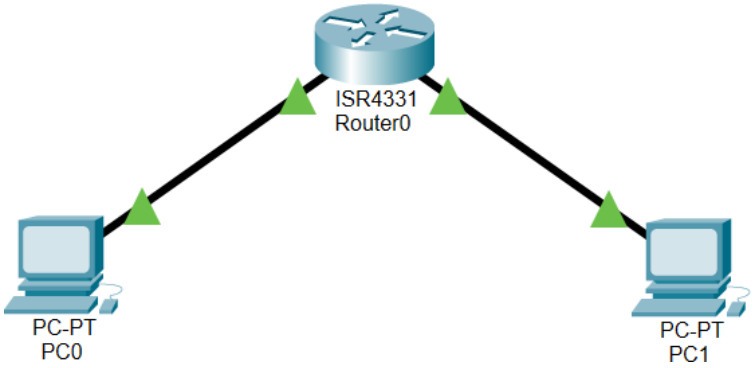
Goal: Connect two PCs on different networks using a router.

Devices:

- 2 PCs
- 1 Router
- 2 Copper Straight-Through Cables

Steps:

1. PC0 → Router Fa0/0, PC1 → Router Fa0/1.



2. Set IPs:
   - PC0: 192.168.1.2 / 255.255.255.0, GW: 192.168.1.1
   - PC1: 192.168.2.2 / 255.255.255.0, GW: 192.168.2.1
3. Configure router:

```
enable
configure terminal
interface fastethernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface fa0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
end
```

Note:

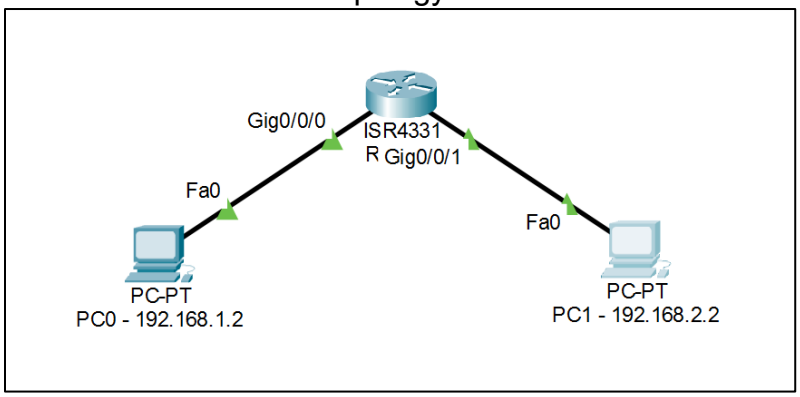
- *You can use int fa 0/0, which also means interface fastethernet 0/0*
- *If the available interface is gigabit, you will use interface gigabitethernet 0/0/0 or int gi 0/0/0*

4. Ping from PC0 to 192.168.2.2

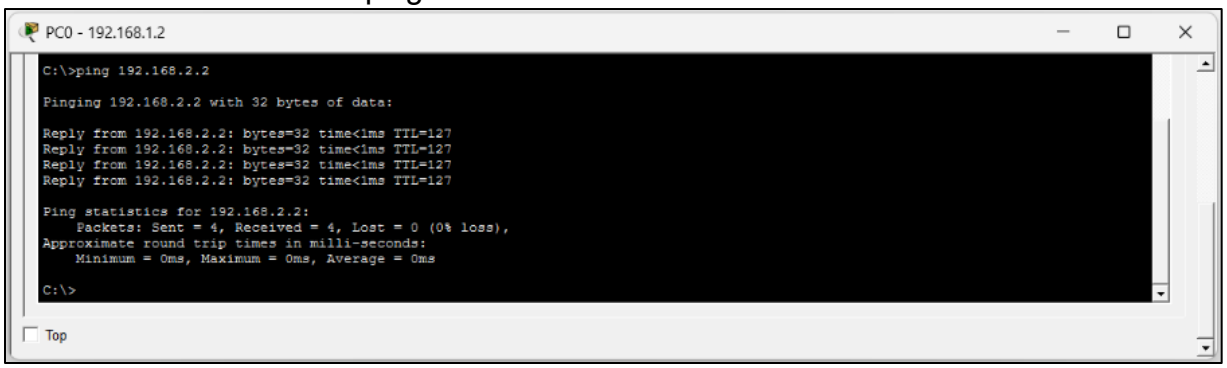
Troubleshooting:

Verify default gateway settings
Ensure router interfaces are no shutdown
Use correct cable type

Documentation:
Screenshot of network topology



Screenshot of successful ping



Notes:
enable → privileged EXEC mode (for higher-level commands configure terminal, show, copy)
configure terminal → global configuration mode. configure hostname, interfaces, routing protocols
interface fa0/0 → Configure Fast Ethernet port
ip address 192.168.1.1 255.255.255.0 → Assign IP Address
no shutdown → To turn it on

TROUBLESHOOTING

- Verify default gateway settings
- Ensure router interfaces are **no shutdown**
- Use correct cable type

GUIDE QUESTIONS

1. What is the main function of a router?
   The main function of a router is to connect different networks and forward data packets between them.

2. Why do routers use IP addresses?
   Routers use IP addresses to identify networks and determine the best path for sending data from one network to another.

Activity 4: VLAN Configuration

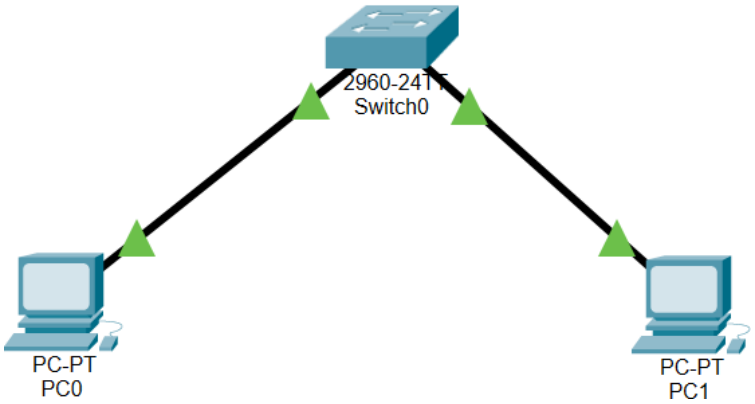
Goal: Implement VLANs to segregate traffic within a switch.

Devices:

- 2 PCs
- 1 Switch

Steps:

1. PC0 → Fa0/1, PC1 → Fa0/2



2. Set IPs:
   - PC0: 192.168.10.1 / 255.255.255.0
   - PC1: 192.168.10.2 / 255.255.255.0
3. Configure VLAN on switch:

```
enable
configure terminal
vlan 10
name Accounting
exit
interface fa0/1
switchport mode access → Access ports belong to only one VLAN, unlike trunk ports
switchport access vlan 10 → Port Membership
exit
interface fa0/2
switchport mode access
switchport access vlan 10
exit
end
```

4. Ping test between PCs.

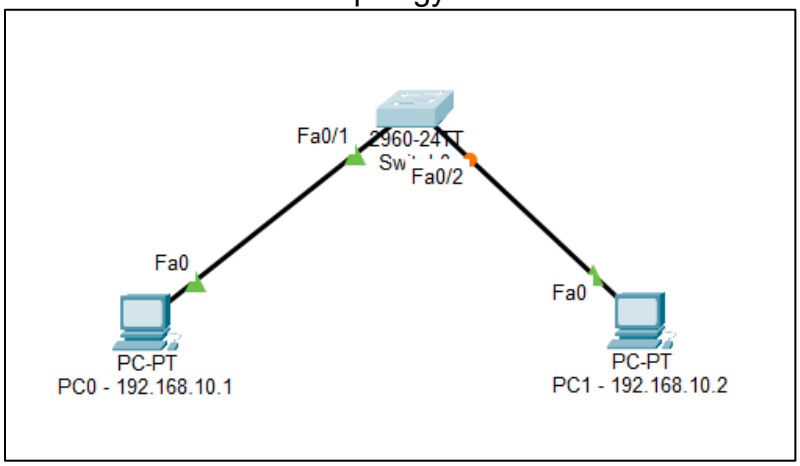
Troubleshooting:

- Ensure that the same VLAN number is assigned.
- Check the switchport mode: it should be set to “access.”
- Make sure the IP address range is correct.

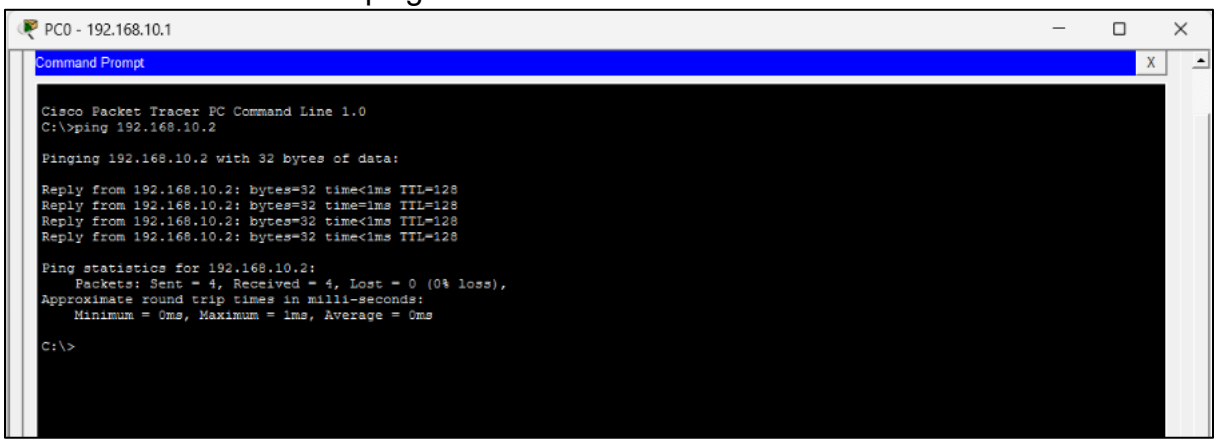
Note: Devices in the same VLAN can communicate. Devices in different VLANs require a router or Layer 3 device.

Documentation:

Screenshot of network topology



Screenshot of successful ping



Activity 5: DHCP Server Configuration

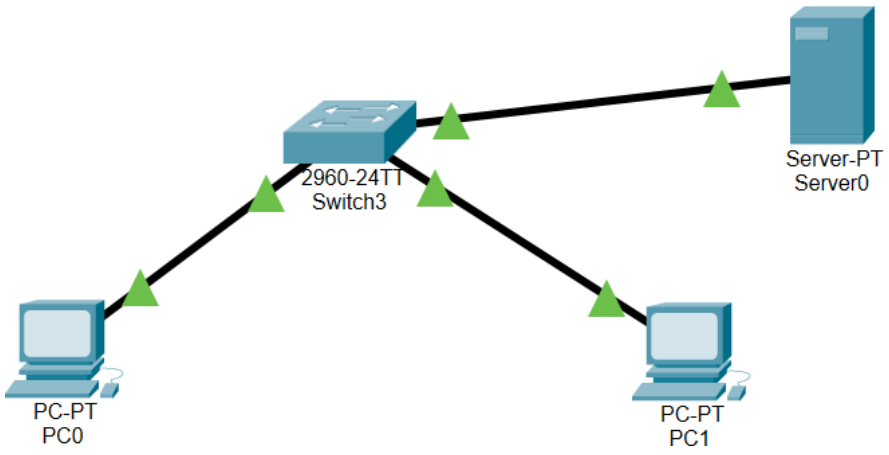
Goal: Automatic IP assignment via DHCP.

Devices:

- 2 PCs
- 1 Switch
- 1 DHCP Server

Steps:

1. Connect all devices to the switch, as shown in the figure below.



2. Configure the DHCP Server > IP Config:
   - IP: 192.168.100.1
   - Subnet Mask: 255.255.255.0
3. Enable and Configure the DHCP Services
   - DHCP Server > Services > DHCP
   - Turn DHCP Services ON
   - Configure the following:
     - Pool Name: PC_Network
     - Default Gateway: 192.168.100.1
     - Start IP: 192.168.100.10
     - Subnet Mask: 255.255.255.0
     - Max Users: 50
4. Configure the PCs:
   - PC > IP Config > Set to DHCP
5. Verify IP Assignment:

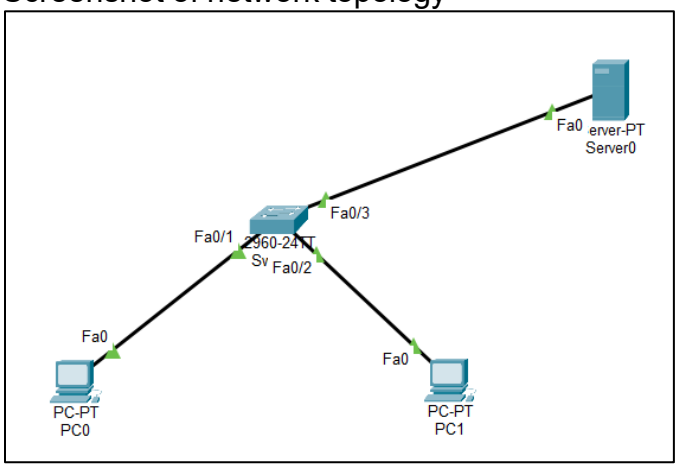
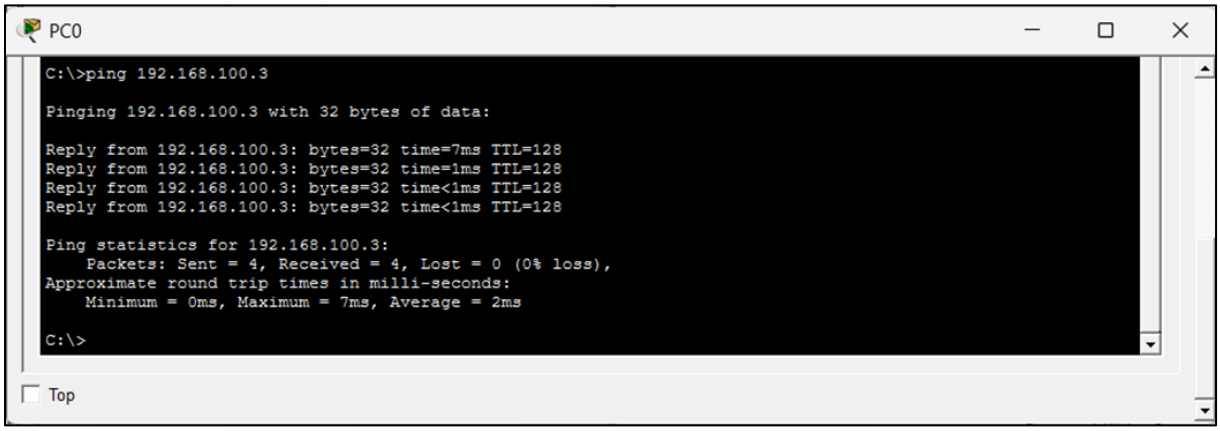
Check if each PC receives an IP address from the DHCP server.

Troubleshooting:

- Ensure that the server is powered on and has the correct IP configuration.
- Verify that the switch is active and properly connected.
- Make sure the PCs are set to DHCP mode.

Documentation:

Screenshot of network topology



Screenshot of successful ping